

GENERAL DISCLOSURE STATEMENT AND CLAIM SCOPE FOUNDATION

The embodiments described throughout this specification are presented for purposes of illustration and explanation and are not intended to limit the scope of the disclosed invention. Individual features, components, processes, or configurations described in connection with specific embodiments may be combined, rearranged, omitted, or substituted without departing from the inventive concepts disclosed herein.

The disclosed intraluminal compliance and distance profiling device may be implemented as a standalone device, as part of a multi-device physiologic measurement system, or as a component within a broader computational ecosystem incorporating sensing nodes, software platforms, artificial intelligence systems, and accessory infrastructures.

Unless explicitly stated otherwise, any element described as singular may be implemented in plural form, and any element described as plural may be implemented in singular form. Structural, functional, or computational components may be distributed across hardware, firmware, software, or remote processing environments while remaining within the scope of the disclosure.

The invention is intended to encompass equivalent structures, technologies, and operational methods capable of achieving substantially similar results using different materials, sensing modalities, algorithms, or system architectures. Future technological developments capable of performing comparable functions are contemplated as equivalents.

Processes described herein may be performed in alternative sequences, concurrently, or through automated or human-assisted workflows. Method steps may be added, removed, or reordered while preserving functional intent.

The disclosure further supports future claims directed to devices, systems, methods of use, computational processing, data analysis, artificial intelligence operation, compatibility modeling, accessory ecosystems, subscription architectures, and platform-level implementations derived from the disclosed concepts.

Nothing in this specification should be interpreted as requiring a particular combination of features unless explicitly recited. Individual features may independently constitute inventive subject matter and may form the basis of future continuation, divisional, or non-provisional patent applications.

The scope of the invention is therefore intended to extend to all variations, modifications, improvements, and equivalents falling within the spirit of the disclosed inventive principles, including embodiments not expressly described but enabled by the teachings contained herein.